



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

VISUAL STUDIO PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Developer Tools

TOTAL: 10 QUESTIONS

**Q1** What is Visual Studio and what problem in Developer Tools does it solve?

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**Q6** How does Visual Studio handle reliability and high availability?

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**Q2** What are the core components or concepts in Visual Studio?

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**Q7** What observability does Visual Studio provide out of the box?

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**Q3** How does Visual Studio compare to its main alternatives?

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**Q8** What are common performance bottlenecks in Visual Studio?

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**Q4** How do you get started with Visual Studio for a new project?

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**Q9** What security best practices apply when running Visual Studio?

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**Q5** What configuration options most affect Visual Studio's behavior in production?

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**Q10** What mistakes do teams commonly make when adopting Visual Studio?

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Visual Studio is a tool or platform

Mitigation

Visual Studio is a tool or platform that automates or simplifies a specific concern in the Developer Tools domain, reducing manual effort and operational complexity for engineering teams.

A6 Visual Studio provides HA through leader election,

Observability

Visual Studio provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Visual Studio has key building blocks that

Primitives

Visual Studio has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Visual Studio exposes metrics (Prometheus endpoint or

Automation

Visual Studio exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Visual Studio trades off simplicity against feature

Hardening

Visual Studio trades off simplicity against feature richness differently than its alternatives. Choose Visual Studio when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Visual Studio via its package manager

Gotchas

Install Visual Studio via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Visual Studio with the principle of

Assessment

Run Visual Studio with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency

Concurrency

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.